

Week 5: Problem Sheet

1. Update Rules

Monte Carlo (MC) Estimation of $V(s)$

The return from state s is:

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots$$

The MC update rule is:

$$V(s) \leftarrow V(s) + \alpha(G_t - V(s))$$

MC methods update value estimates using complete episodes.

Temporal Difference TD(0)

The TD target is:

$$R_{t+1} + \gamma V(s_{t+1})$$

The TD(0) update rule is:

$$V(s_t) \leftarrow V(s_t) + \alpha[R_{t+1} + \gamma V(s_{t+1}) - V(s_t)]$$

TD methods update values at every step using bootstrapping.

SARSA (On-policy TD Control)

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[R_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)]$$

Q-learning (Off-policy TD Control)

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[R_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

2. Monte Carlo vs Temporal Difference

Aspect	Monte Carlo	Temporal Difference
Updates	End of episode	Every step
Bootstrapping	No	Yes
Variance	High	Lower
Sample efficiency	Low	High
Convergence	Slower	Faster in practice

TD methods are more sample efficient and converge faster than Monte Carlo methods.